

Business Problem Statement

Customer Trends Analysis using SQL, Python & Power BI

Background

A retail company wants to better understand its customers' shopping behavior to improve business performance and customer satisfaction. The company collects customer transaction and demographic data but has not fully utilized this information to support business decisions.

By analyzing customer shopping patterns, sales performance, and purchasing preferences, the company aims to identify opportunities for growth, improve marketing strategies, and enhance the overall customer experience.

Business Objective

Analyze customer shopping data using SQL, Python, and Power BI to identify customer trends, understand purchasing behavior, and generate actionable business insights that support data-driven decision-making.

Business Questions

- Which customer segments generate the highest sales?
- Which product categories perform the best?
- How do age and gender influence purchasing behavior?
- Which sales channels are preferred by customers?
- How do discounts impact customer purchases?
- Which payment methods are most commonly used?
- Which regions contribute the highest revenue?
- What are the monthly sales trends?
- How can the company improve customer engagement and retention?

Key Performance Indicators (KPIs)

- Total Sales
- Total Orders
- Average Order Value
- Customer Purchase Frequency
- Product Category Performance

- Regional Sales
- Sales Channel Performance
- Payment Method Distribution
- Discount Usage
- Customer Demographics

Project Deliverables

1. Data Preparation (Python): Clean and prepare the dataset for analysis.
2. Data Analysis (SQL): Analyze customer behavior and business metrics using SQL queries.
3. Interactive Dashboard (Power BI): Build a dashboard to visualize KPIs and trends.
4. Business Insights & Recommendations: Summarize findings and provide recommendations.
5. GitHub Repository: Organize all project files and documentation.

Expected Outcomes

- Understand customer purchasing behavior.
- Identify high-performing products and customer segments.
- Monitor sales performance across regions and channels.
- Improve marketing strategies.
- Support business decisions with data insights.

Tools & Technologies

SQL, Python, Pandas, NumPy, Power BI, Microsoft Excel, Git & GitHub